Motor wiring card PCA9685 → TB6612FNG · robocar

Six signal wires, plus one that does not come from the PCA9685
Channels 8–13 carry direction and speed for both motors. **STBY** is the seventh wire and runs straight from the XIAO's **D0 / GPIO1** — it is not a PCA9685 channel. With STBY low the H-bridges stay disabled and the motors do nothing, however correct the other six wires are.

1 · Board layouts, viewed from the component side

PCA9685 16-channel PWM driver — Adafruit #815 and its clones

V+ **GND** screw terminal — servo/motor rail in

GND
OE
SCL
SDA
VCC
V+

PCA9685
16 × 12-bit PWM · addr 0x40 · 100 Hz

I²C address jumpers A0–A5 along this top edge

GND
OE
SCL
SDA
VCC
V+

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
PWM	PWM	PWM	PWM	PWM	PWM	PWM	PWM	R PWM	R IN2	R IN1	L IN1	L IN2	L PWM	PWM	PWM
V+	V+	V+	V+	V+	V+	V+	V+	V+	V+	V+	V+	V+	V+	V+	V+
GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND

Rows, board edge inwards: **GND** (outermost) · **V+** · **PWM** (nearest the chip). Channels run 0 on the left to 15 on the right, in four blocks of four. Only the **PWM** row of the marked columns is wired to the motor driver.

Both side headers carry the same six pins
The 6-pin headers on the left and right edges are identical — GND, OE, SCL, SDA, VCC, V+ top to bottom — so either one can take the I²C feed and the other chains onward. Here the I²C side comes from **TCA9548A channel 0**, not from the XIAO directly. The board's own underside warns: **the screw terminal is reverse-polarity protected, the side breakout pins are not** — the protection MOSFET sits behind the terminal only, so V+ fed backwards through a side header kills the board. **OE** needs nothing: a 10 kΩ pulldown on the board already enables the outputs.

TB6612FNG dual motor driver — SparkFun ROB-14451 and its clones

top of board — SparkFun flame logo at this end

VM
VCC
GND
A01
A02
B02
B01
GND

TB6612FNG
dual H-bridge · 1.2 A avg / 3.2 A peak

PWMA
AIN2
AIN1
STBY
BIN1
BIN2
PWMB
GND

from PCA9685 · from XIAO D0 · rails · to motors · unused

Both 8-pin rows start at the same end: **VM** and **PWMA** are the top corners, both rows end in **GND**. Outputs are on the VM side, control on the PWMA side. Pins are labelled on both faces; the second GND is redundant on-board.

A01 / A02 / B01 / B02 are printed with a zero, not a letter O
Channel **A drives the right motor** and **B the left**, matching `docs/schematics/circuits/robocar.py` and the `MOTOR_RIGHT_*` / `MOTOR_LEFT_*` names in `main/pin_config.h`. Nothing in the firmware can tell which motor is which — that is set entirely by which output pair you solder to which wheel.

2 · The seven wires

Rows are in the motor driver's own pin order, top to bottom, so working down this table is working down the header. Channel numbers ascend with it — that is the point of the assignment, not a coincidence.

	TB6612 pin	From	Carries
□	1 · **PWMA**	PCA9685 ch 8 · PWM row	Right motor speed — the only modulated line of the three
□	2 · **AIN2**	PCA9685 ch 9 · PWM row	Right motor direction bit 2
□	3 · **AIN1**	PCA9685 ch 10 · PWM row	Right motor direction bit 1 — driven full-on (4096) or full-off (0), never a duty cycle
□	4 · **STBY**	XIAO **D0 / GPIO1**	Global enable, straight from the MCU — the one wire in this column that is not a PCA9685 channel
□	5 · **BIN1**	PCA9685 ch 11 · PWM row	Left motor direction bit 1
□	6 · **BIN2**	PCA9685 ch 12 · PWM row	Left motor direction bit 2
□	7 · **PWMB**	PCA9685 ch 13 · PWM row	Left motor speed
—	8 · GND	—	Redundant on-board with the other GND; leave it

The A side is PWM-then-direction, the B side direction-then-PWM
Not an inconsistency — the header is symmetric about STBY rather than repeated, so following it in order is what removes the crossings. If the channel numbers ever stop ascending down this table, the wiring and `main/pin_config.h` have diverged.

Leave the V+ and GND rows of those six columns unconnected
Each PCA9685 channel breaks out three pins, but the motor driver takes only the signal. The V+ and GND pins of columns 8–13 stay empty; the driver gets its power and its ground from the rails below, not from the PWM header.

3 · Power and ground

	Pin	Rail	Note
□	TB6612 **VM**	5.0 V from the LM2596	Motor supply. Star-wire it from the regulator's output terminal — do not daisy-chain off the amplifier or the servo feed
□	TB6612 **VCC**	3.3 V	Logic supply. Must match the PCA9685's VCC — see §4
□	PCA9685 **VCC**	3.3 V	Logic supply, via the side header
□	PCA9685 **V+** (screw terminal)	5.0 V from the LM2596	Servo rail. 6 V absolute maximum, per the board's own silkscreen. Fit ≥470 µF here
□	TB6612 **GND** (either pin)	Common ground	Both GND pins are joined on-board; one wire is enough
□	PCA9685 **GND**	Common ground	Terminal-block GND and header GND are the same net

A missing ground does not stop a chip from running
A board whose GND is not connected finds a return path through the ESD clamp diodes on its signal pins and keeps working — badly, in a way that reads as a firmware bug. Buzz every ground to the regulator's negative terminal **before** applying power, not after something misbehaves.

4 · Why both logic rails are 3.3 V

Both parts specify their input threshold as a fraction of **their own** supply, so the rail is not a free choice:

Part	Spec	Consequence
PCA9685 SCL/SDA	$V_{IH} = 0.7 \times V_{DD}$	At V_{DD} = 5 V the threshold is 3.5 V, above what the XIAO's 3.3 V I²C can drive. At 3.3 V it is 2.31 V

Part	Spec	Consequence
TB6612 IN1/IN2/PWM	$V_{IH} = 0.7 \times V_{CC}$	At V_{CC} = 5 V the threshold is 3.5 V, above the 3.3 V the PCA9685 would output
TB6612 STBY	$V_{IH} = 0.7 \times V_{CC}$	At V_{CC} = 5 V a 3.3 V GPIO cannot reliably lift STBY — the motors would stay in standby

Only VM and V+ take 5 V
The 5 V rail reaches TB6612 **VM** and PCA9685 **V+** and nothing else on this card; both VCC pins are logic supplies and take 3.3 V. `WIRING.md` and the schematic at `docs/schematics/circuits/robocar.py` agree on this.

5 · Before power-up, and first motion

	Check	Expected
☐	Continuity, every GND pin to the regulator's negative terminal	Beeps — all of them
☐	Continuity, TB6612 VM to TB6612 VCC	**No** beep. A beep means the 5 V and 3.3 V rails are shorted
☐	Regulator output, unloaded, before anything is connected	5.0 V — the LM2596 trimpot spans 1.2–37 V and arrives set at neither
☐	`servo` on the console	Reports the live PWM frequency — proves the PCA9685 answers on TCA9548A ch 0 before any motor moves
☐	`F` on the console, wheels off the ground	Both wheels forward for about a second, then stop
☐	`trace` on the console	I²C op counter climbing while driving, near-idle when stopped

Truth table, per channel

IN1	IN2	PWM	Mode
H	L	H	CW
L	H	H	CCW
H	L	L	Short brake
H	H	—	Short brake
L	L	H	Stop (coast)

STBY must be high for any of it. The firmware drives forward as IN1 = high, IN2 = low.

If a wheel runs backwards

Swap that motor's **two output wires** at A01/A02 or B01/B02. Do not swap the IN1/IN2 wires and do not change the firmware: the direction bits are shared with the reactive controller's turn logic, so inverting them there fixes driving straight and breaks turning.

If nothing moves at all

Measure STBY. Below 2.31 V with V_{CC} at 3.3 V, the bridges are held off and every other wire is irrelevant.

Sources. Channel numbers and the STBY GPIO: `main/pin_config.h` via `docs/auto/pin_defs.typ`, regenerated and diffed by `.github/workflows/build-guide-check.yml`. Board layouts are read out of the vendors' published Eagle files — Adafruit `Adafruit-16-Channel-PWM-Servo-Driver-PCB` (`Adafruit PCA9685 rev C.brd`) and SparkFun `Motor_Driver-Dual_TB6612FNG` (`SparkFun_Motor_Driver-TB6612FNG_v11.brd`) — not from a product photo. Thresholds: NXP PCA9685 datasheet rev. 4 and Toshiba TB6612FNG datasheet, both listed in `docs/reference/datasheets/`. Rails and decoupling: `WIRING.md` § Power.